

BILL TRUONG

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PROFESSIONAL SUMMARY

Unity engineer and technical artist. I build gameplay systems and the graphics tech behind them: custom shaders in HLSL, toon and cel-shading, VFX, and render optimization for platforms as tight as the Meta Quest. I have shipped VR, mobile, and hyper-casual titles, and built a Unity-to-Omniverse digital-twin pipeline that was shown to NVIDIA CEO Jensen Huang.

TECHNICAL SKILLS

Languages: C# (Advanced), Python (tooling/automation), HLSL/Cg (Advanced), C++ , JavaScript, SQL

Unity Engine: Unity 2022/6 LTS, DOTS (ECS/Jobs), URP, HDRP, Editor Scripting, Addressables, Profiling (Deep Profile, Memory Profiler).

Graphics & Tech Art: Custom Shaders (HLSL/Shader Graph), VFX Graph, Compute Shaders, GPU Instancing, Render Graph API, Raymarching, Post-Processing Stack.

Backend & Networking: Photon Fusion, SmartFox Server, RESTful APIs, Firebase, Real-time Replication, Multi-threading.

Tools & Platforms: NVIDIA Omniverse, Meta Quest SDK, Blender (Python API), RenderDoc, Git/GitLab, Jira, Confluence, Figma, Adobe Substance Painter.

PROFESSIONAL EXPERIENCE

Curly Blue (Shmackle VR)

Ho Chi Minh City, Vietnam

Unity Developer, promoted to Technical Artist

Apr 2025 – Present

- Rebuilt the visual style of *Shmackle VR*, replacing flat unlit assets with a custom toon and cel-shading look. Wrote the shaders in HLSL and Shader Graph, and added the VFX and dissolve effects.
- Kept the new visuals from costing performance by cutting draw calls, using GPU instancing, and atlasing textures. Held the 72 FPS baseline on Quest 2 and reached roughly 80 to 90 FPS on Quest 3.
- Built the multiplayer layer on Photon Fusion, syncing physics and game state for up to 8 concurrent players in the social sandbox.
- Added a collider-triggered activation system that turns objects on and off as players enter or leave a zone, which lowered render load.
- Shipped the "Shooting Range" update (modular weapon system, target pooling) and the diegetic "Shmackle Pad" UI, along with a dynamic FOV adjustment to reduce motion sickness. Active-user counts rose across successive releases.

FPT IS (akaVerse)

Ho Chi Minh City, Vietnam

Software Engineer, R&D

Jul 2024 – Apr 2025

- Built a real-time two-way sync pipeline between Unity and NVIDIA Omniverse for an industrial digital-twin product, made at FPT for a client in Japan. The product was shown at the NVIDIA summit and presented directly to CEO Jensen Huang.
- Worked as a generalist across the studio: wrote internal Python tools, profiled web performance, and mentored interns on 3D workflows. Was one of the highest-rewarded engineers in the southern office that year.
- Wrote "Bill Utils," an in-house Python tool for the heritage 3D-scanning pipeline. The government project could not use outside software, so I built a mesh-cleanup utility that strips redundant geometry from scanned woodblocks and cuts the manual work artists had to do by hand.
- Core Developer on *Animal Push Royale*, a GameVerse 2024 finalist. Tripled the framerate on low-end devices, from a rendering-bound 30 FPS to a stable 90, by removing GC allocations and cleaning up hot render loops.
- Used Compute Shaders to render large CAD datasets in WebGL, working around browser memory limits for enterprise clients.

Golden Sea Studio

Ho Chi Minh City, Vietnam

Game Developer

2024

- Built and shipped hyper-casual titles on one-month cycles, taking games like *Merge Fruit* from concept to release.
- Added third-party SDKs for ads and analytics, and trimmed build sizes to help with user acquisition.

Gameloft

Ho Chi Minh City, Vietnam

Gameplay Programmer

Jun 2023 – Dec 2023

- Worked on *Applaydu* (50M+ downloads) inside a large international team, following Agile/Scrum.
- Wrote gameplay for the "Treasure Toss" and "Animalody" modules, using Observer, Command, and Singleton patterns to keep the code modular and testable.
- Iterated on game feel with designers and QA to meet the licensor's (Ferrero) quality standards.

KEY PROJECTS & OPEN SOURCE

- Pro Origin Tools**  | *Commercial Blender Add-on (Superhive / Blender Market)* Python, bpy
- Built and sell a Blender add-on for smart origin handling. It detects separate mesh islands, so each leg of a joined chair gets its own candidate origin, and adds a one-click hinge tool for doors and lids.
 - Runs every mode across a whole selection in batch, and exposes a headless bpy.ops API for pipeline scripts. Ships with Unity, Unreal, and Godot ground presets; pure Python across Blender 4.2 to 5.1, no dependencies.
- Kindly's Promise (Project Anchor)**  | *Personal Indie Game* Unity URP, C#, HLSL
- Built a vertical slice of a narrative sci-fi adventure game as the solo developer and technical artist.
 - Wrote a custom render setup with a see-through dissolve effect using stencil buffers, plus a dynamic cel-shading solution, and a modular interaction system with a non-linear dialogue engine.
- KeyStream-Gemini**  | *Open Source (38 stars, 14 forks)* JavaScript
- Built a reverse proxy for the Google Gemini API that rotates across multiple keys and monitors them in real time, keeping requests flowing when any single key hits its rate limit.
- Bill SSOOutline (URP)**  | *Graphics Programming* Render Graph API
- Wrote a screen-space outline render feature on URP's Render Graph system.
 - Fixed the depth-occlusion artifacts that a plain Sobel pass produces by adding depth-normal sensitivity checks.
- Lucy**  | *Self-hosted Personal AI Agent* Claude Agent SDK, Node.js, SQLite
- Built an AI agent that runs 24/7 on a 1.9 GB VPS and talks over Telegram, with most of the work on long-term memory. Three tiers run live (full-text FTS5 recall, hybrid keyword-plus-vector search fused with RRF, and episodic per-turn storage); self-consolidation and bi-temporal forgetting sit behind a dry-run gate.
 - Added a tool registry with MCP (files, web, git), a guard that blocks dangerous commands, and an auto-task engine that plans, builds, and reports on its own.
- Other Open Source**  | *Shaders & Tools* HLSL, C#, Python
- Bill Biome Shader (stylized environment suite with an editor painting tool, GPU instancing), Pro Atlas Baker (Blender atlasing/UV packing), Scene Switcher (Unity editor tool), and a Responsive WebGL Template.

LEADERSHIP & COMMUNITY

- FullHouseDev** Remote
Game Development Instructor Sep 2024 – Present
- Teach Unity architecture and shader development, turning engineering topics into lessons aspiring developers can follow.
- Content Creator (Bill The Dev)** YouTube, LinkedIn
Technical Writer & Educator 2023 – Present
- Run a blog and video channel on Unity optimization and graphics programming.
 - Contribute open-source tools and templates to the game-dev community.

EDUCATION

- FPT University** Ho Chi Minh City, Vietnam
Bachelor of Software Engineering 2022 – 2025
- Talent Scholarship: 30% tuition, awarded for placing in the top 30% of 10,000+ candidates.
 - Relevant coursework: Data Structures & Algorithms, Computer Graphics, Software Architecture, 3D Mathematics.
- EMASI Van Phuc International School** Ho Chi Minh City
High School Diploma (Honors), GPA 9.2/10 2019 – 2022

CERTIFICATIONS & AWARDS

- Unity:** Complete C# Unity Game Developer 2D (GameDev.tv / Udemy)
Apple: Selected by FPT IS for developer training led by Apple engineers
Language: IELTS 6.0
Award: GameVerse 2024 Finalist (Animal Push Royale)